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Anomie's SPC700 Cycle Doc

\$Revision: 1126 \$

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1 Register, Immediate -- A,#i; X,#i; Y,#i
(ADC,AND,CMP,CMP,CMP,EOR,MOV,MOV,MOV,OR,SBC)
(2 bytes)
(2 cycles)

1	PC	Op Code	1
2	PC+1	Data	1

* This should be accurate.

2 Register, Register -- A,X; A,Y; X,A; X,Y; Y,A; Y,X; SP,X; X,SP
(MOV,MOV,MOV,MOV,MOV,MOV)
(1 byte)
(2 cycles)

1	PC	Op Code	1
2	??	I/O	?

* This should be accurate.

3 Register, Direct -- A,d; X,d; Y,d
(ADC,AND,CMP,CMP,CMP,EOR,MOV,MOV,MOV,OR,SBC)
(2 bytes)
(3 cycles)

1	PC	Op Code	1
2	PC+1	D0	1
3	D0	Data	1

* Verified by blargg.

4 Register, Direct Indexed -- A,d+X; X,d+Y; Y,d+X
(ADC,AND,CMP,EOR,MOV,MOV,MOV,OR,SBC)
(2 bytes)
(4 cycles)

1	PC	Op Code	1
2	PC+1	D0	1
3	??	I/O	?
4	D0	Data	1

* blargg verified Data read is cycle 4.

* 2 and 3 could be swapped, but that would be odd.

5a Register, Indirect -- A,(X)
(ADC,AND,CMP,EOR,MOV,OR,SBC)
(1 byte)
(3 cycles)

1	PC	Op Code	1
2	??	I/O	?
3	X	Data	1

* Verified by blargg.

5b Register, Indirect++ -- A,(X)+
(MOV)
(1 byte)
(4 cycles)

1	PC	Op Code	1
2	??	I0	?
3	X	Data	1
4	??	I0	?

* Verified by blargg.

6 Register, Indexed Indirect -- A,[d+X]
(ADC,AND,CMP,EOR,MOV,OR,SBC)
(2 bytes)
(6 cycles)

1	PC	Op Code	1
2	PC+1	D0	1
3	??	I0	?
4	D0+X	AAL	1
5	D0+X+1	AAH	1
6	AA	Data	1

* blargg verifies the Data read is cycle 6.

* Cycles 2-5 could be rearranged, but this is most likely.

7 Register, Indirect Indexed -- A,[d]+Y
(ADC,AND,CMP,EOR,MOV,OR,SBC)
(2 bytes)
(6 cycles)

1	PC	Op Code	1
2	PC+1	D0	1
3	D0	AAL	1
4	D0+1	AAH	1
5	??	I0	?
6	AA+Y	Data	1

* blargg verifies the Data read is cycle 6.

* Cycles 2-5 could be rearranged, but this is most likely.

8 Register, Absolute -- A,!a; X,!a; Y,!a
(ADC,AND,CMP,CMP,CMP,EOR,MOV,MOV,MOV,OR,SBC)
(3 bytes)
(4 cycles)

1	PC	Op Code	1
2	PC+1	AAL	1
3	PC+2	AAH	1
4	AA	Data	1

* Verified by blargg.

* 2 and 3 could be swapped, but that would be odd.

9 Register, Absolute Indexed -- A,!a+X; A,!a+Y
(ADC,ADC,AND,AND,CMP,CMP,EOR,EOR,MOV,MOV,OR,OR,SBC,SBC)
(3 bytes)
(5 cycles)

1	PC	Op Code	1
2	PC+1	AAL	1
3	PC+2	AAH	1

4	??	I0	?
5	AA+X/Y	Data	1

* blargg verifies Data read is cycle 5.
 * Cycles 2-4 could be in other orders, but this is most likely.

10 Direct, Immediate -- d,#i
 (ADC,AND,CMP,EOR,MOV,OR,SBC)
 (3 bytes)
 (5 cycles)

1	PC	Op Code	1
2	PC+1	Data 1	1
3	PC+2	D0	1
4	D0	Data 2 (read)	1
5	D0	Data 2 (write)	0

* Verified by blargg. 2 and 3 could be swapped, but that's unlikely.
 * Yes, RMW even for MOV.
 * CMP does not write for cycle 5; does it I0 or read again?

11 Direct, Register -- d,A; d,X; d,Y
 (MOV,MOV,MOV)
 (2 bytes)
 (4 cycles)

1	PC	Op Code	1
2	PC+1	D0	1
3	D0	Data (read)	1
4	D0	Data (write)	0

* Verified by blargg.
 * Yes, RMW even for MOV

12 Direct Indexed, Register -- d+X,A; d+X,Y; d+Y,X
 (MOV,MOV,MOV)
 (2 bytes)
 (5 cycles)

1	PC	Op Code	1
2	PC+1	D0	1
3	??	I0	1
4	D0+X/Y	Data (read)	1
5	D0+X/Y	Data (write)	0

* Verified by blargg. 2 and 3 could be swapped, but that's unlikely.
 * Yes, RMW even for MOV

13a Indirect, Register -- (X),A
 (MOV)
 (1 byte)
 (4 cycles)

1	PC	Op Code	1
2	??	I0	?
3	X	Data (read)	1
4	X	Data (write)	0

* Verified by blargg.
 * Yes, RMW even for MOV

13b Indirect++, Register -- (X)+,A

(MOV)

(1 byte)

(4 cycles)

1	PC	Op Code	1
2	??	I0	?
3	??	I0	?
4	D0	Data (write)	0

* Verified by blargg.

* No RMW here

14 Indexed Indirect, Register -- [d+X],A

(MOV)

(2 bytes)

(7 cycles)

1	PC	Op Code	1
2	PC+1	D0	1
3	??	I0	?
4	D0+X	AAL	1
5	D0+X+1	AAH	1
6	AA	Data (read)	1
7	AA	Data (write)	0

* blargg verifies cycles 6 and 7.

* Yes, RMW even for MOV

15 Indirect Indexed, Register -- [d]+Y,A

(MOV)

(2 bytes)

(7 cycles)

1	PC	Op Code	1
2	PC+1	D0	1
3	D0	AAL	1
4	D0+1	AAH	1
5	??	I0	?
6	AA+Y	Data (read)	1
7	AA+Y	Data (write)	0

* blargg verifies cycles 6 and 7.

* Yes, RMW even for MOV

16 Absolute, Register -- !a,A; !a,X; !a,Y

(MOV,MOV,MOV)

(3 bytes)

(5 cycles)

1	PC	Op Code	1
2	PC+1	AAL	1
3	PC+2	AAH	1
4	AA	Data (read)	1
5	AA	Data (write)	0

* Verified by blargg. 2 and 3 could be swapped, but that's unlikely.

* Yes, RMW even for MOV

17 Absolute Indexed, Register -- !a+X,A; !a+Y,A

(MOV,MOV)

(3 bytes)
(6 cycles)

1	PC	Op Code	1
2	PC+1	AAL	1
3	PC+2	AAH	1
4	??	I0	?
5	AA+X/Y	Data (read)	1
6	AA+X/Y	Data (write)	0

* blargg verified cycles 5 and 6.
* Yes, RMW even for MOV

18 Direct, Direct -- dd,ds
(ADC,AND,CMP,EOR,MOV,OR,SBC)
(3 bytes)
(5 or 6 cycles)

1	PC	Op Code	1
2	PC+1	DS	1
3	DS	Data 1	1
4	PC+2	DD	1
[5]	DD	Data 2 (read)	1
6	DD	Data 2 (write)	0

* Verified by blargg.
* Skip cycle 5 for MOV, no RMW.
* CMP does not write for cycle 6; does it I0 or read again?

19 Indirect, Indirect -- (X),(Y)
(ADC,AND,CMP,EOR,OR,SBC)
(1 byte)
(5 cycles)

1	PC	Op Code	1
2	??	I0	?
3	Y	Data 1	1
4	X	Data 2 (read)	1
5	X	Data 2 (write)	0

* Verified by blargg.
* CMP does not write for cycle 5; it is an I0 cycle.

20a Direct (RMW) -- d
(ASL,CLR1,DEC,INC,LSR,R0L,R0R,SET1)
(2 bytes)
(4 cycles)

1	PC	Op Code	1
2	PC+1	D0	1
3	D0	Data (read)	1
4	D0	Data (write)	0

* Verified by blargg.

20b Direct (RMW) -- m.b
(NOT1)
(3 bytes)
(5 cycles)

1	PC	Op Code	1
2	PC+1	AAL	1
3	PC+2	AAH & BIT	1

4	D0	Data (read)	1
5	D0	Data (write)	0

* Verified by blargg. 2 and 3 could be swapped, but that's unlikely.

21 Direct Indexed (RMW) -- d+X
(ASL,DEC,INC,LSR,ROL,ROR)
(2 bytes)
(5 cycles)

1	PC	Op Code	1
2	PC+1	D0	1
3	??	I0	?
4	D0	Data (read)	1
5	D0	Data (write)	0

* blargg verified cycles 4 and 5.

* 2 and 3 could be swapped, but that would be odd.

22 Absolute (RMW) -- !a
(ASL,DEC,INC,LSR,ROL,ROR,TCLR1,TSET1)
(3 bytes)
(5 or 6 cycles)

1	PC	Op Code	1
2	PC+1	AAL	1
3	PC+2	AAH	1
[4]	AA	Data (read)	1
5	AA	Data (read)	1
6	AA	Data (write)	1

* Verified by blargg.

* 2 and 3 could be swapped, but that would be odd.

* Cycle 4 only for TSET1 and TCLR1.

23 Carry, MemBit -- C,m.b; C,/m.b
(AND1,AND1,MOV1,EOR1,OR1,OR1)
(3 bytes)
(4 or 5 cycles)

1	PC	Op Code	1
2	PC+1	AAL	1
3	PC+2	AAH & BIT	1
4	AA	Data	1
[5]	??	I0	?

* Verified by blargg.

* Cycle 5 present only in EOR1 and OR1.

24 MemBit, Carry -- m.b,C
MOV1 m.b, C CA 3 6 (m.b) =

C

(MOV1)
(3 bytes)
(6 cycles)

1	PC	Op Code	1
2	PC+1	AAL	1
3	PC+2	AAH & BIT	1
4	AA	Data (read)	1
5	??	I0	?

6 AA Data (write) 0
 * Verified by blargg. 2 and 3 could be swapped, but that's unlikely.

25 Implied

(ASL, CLRC, CLRP, CLRV, DAA, DAS, DEC, DEC, DEC, DI, DIV, EI, INC, INC, INC, LSR, MUL, NOP,

NOTC, ROL, ROR, SETC, SETP, XCN)

(1 byte)

(2, 3, 5, 9, or 12 cycles)

1	PC	Op Code	1
2	??	I0	?
[3-12]	??	I0	?

* This should be accurate.

* The common theme here is that all these do no memory accesses.

26 Word-Register, Direct -- YA,d

(ADDW, CMPW, MOVW, SUBW)

(2 bytes)

(4 or 5 cycles)

1	PC	Op Code	1
2	PC+1	D0	1
3	D0	Data 1 Low	1
[4]	??	I0	?
5	D0+1	Data 1 High	1

* Verified by blargg.

* No cycle 4 for CMPW

27 Direct, Word-Register, Direct -- d,YA

(MOVW)

(2 bytes)

(5 cycles)

1	PC	Op Code	1
2	PC+1	D0	1
3	D0	Data Low (read)	1
4	D0	Data Low	0
5	D0+1	Data High	0

* Verified by blargg.

* Yes, RMW on low byte only...

28 Word-Direct (RMW) -- d

(DECW, INCW)

(2 bytes)

(6 cycles)

1	PC	Op Code	1
2	PC+1	D0	1
3	D0	Data 1 Low	1
4	D0	Data 1 Low	0
5	D0+1	Data 1 High	1
6	D0+1	Data 1 High	0

* Verified by blargg.

29 Stack Ops

(POP,POP,POP,POP,PUSH,PUSH,PUSH,PUSH)
 (1 byte)
 (4 cycles)

1	PC	Op Code	1
2	??	I0	?
3	S	Data	0/1
4	??	I0	?

* Or is it Data-I0-I0 or I0-I0-Data?

30a JMP Absolute -- !a

(JMP)
 (3 bytes)
 (3 cycles)

1	PC	Op Code	1
2	PC+1	AAL	1
3	PC+2	AAH	1
(1)	New PC	Op Code	1

* This should be accurate.

30b JMP Absolute Indexed Indirect -- [!a+X]

(JMP)
 (3 bytes)
 (6 cycles)

1	PC	Op Code	1
2	PC+1	AAL	1
3	PC+2	AAH	1
4	??	I0	?
5	AA	PCL	1
6	AA+1	PCH	1
(1)	New PC	Op Code	1

* This should be fairly accurate.
 * Any penalty for AA+X crossing pages?

31a Branch -- r

(BCC,BCS,BEQ,BMI,BNE,BPL,BVC,BVS,BRA)
 (2 bytes)
 (2 or 4 cycles)

1	PC	Op Code	1
2	PC+1	R	1
[3]	??	I0	?
[4]	??	I0	?

* This should be accurate.
 * Cycles 3 and 4 only present if branch is taken.
 * BRA always takes the branch.

21b Test-and-Branch Direct -- d,r; d.b,r

(BBC,BBS,CBNE)
 (3 bytes)
 (5 or 7 cycles)

1	PC	Op Code	1
2	PC+1	D0	1
3	D0	Data	1
4	PC+2	R	1

5	??	I0	?
[6]	??	I0	?
[7]	??	I0	?

* Cycles 6 and 7 only present if branch is taken

* Cycles 1-3 are verified by blargg.

31c Test-and-Branch Direct Indexed -- d+X,r

(CBNE)

(3 bytes)

(6 or 8 cycles)

1	PC	Op Code	1
2	PC+1	D0	1
3	??	I0	?
4	D0+X	Data	1
5	PC+2	R	1
6	??	I0	?
[7]	??	I0	?
[8]	??	I0	?

* blargg verifies Data read is cycle 4; the rest is guesswork.

* Order of cycles is unknown, except the two extras are probably
are at the
end.

31d Modify-and-Branch Direct -- d,r; d.b,r

(DBNZ)

(3 bytes)

(5 or 7 cycles)

1	PC	Op Code	1
2	PC+1	D0	1
3	D0	Data (read)	1
4	D0	Data-1 (write)	0
5	PC+2	R	1
[6]	??	I0	?
[7]	??	I0	?

* blargg verifies cycles 3 and 4.

* Cycles 6 and 7 only present if branch is taken

* Order of cycles is unknown, except the two extras are probably
are at the
end.

31e Modify-and-Branch Register -- Y,r

(DBNZ)

(2 bytes)

(4 or 6 cycles)

1	PC	Op Code	1
2	PC+1	R	1
3	??	I0	?
4	??	I0	?
[5]	??	I0	?
[6]	??	I0	?

* Cycles 5 and 6 only present if branch is taken

* Order of cycles is unknown, except the two extras are probably
are at the
end.

32a CALL
(CALL)
(3 bytes)
(8 cycles)

1	PC	Op Code	1
2	SP	PCH	0
3	SP-1	PCL	0
4	??	I0	?
5	PC+1	AAL	1
6	PC+2	AAH	1
7	??	I0	?
8	??	I0	?
(1)	new PC	Op Code	1

* WTF with all the I0 cycles?

* Order of reading new addr and pushing old addr may be wrong.

32b PCALL
(PCALL)
(2 bytes)
(6 cycles)

1	PC	Op Code	1
2	SP	PCH	0
3	SP-1	PCL	0
4	PC+1	U	1
5	Vec	AAL	1
6	Vec+1	AAH	1
(1)	new PC	Op Code	1

* Order of reading new addr and pushing old addr may be wrong.

32c TCALL
(TCALL)
(1 byte)
(8 cycles)

1	PC	Op Code	1
2	SP	PCH	0
3	SP-1	PCL	0
4	??	I0	?
5	Vec	AAL	1
6	Vec+1	AAH	1
7	??	I0	?
8	??	I0	?
(1)	new PC	Op Code	1

* WTF with all the I0 cycles?

* Order of reading new addr and pushing old addr may be wrong.

32d BRK
(BRK)
(1 byte)
(8 cycles)

1	PC	Op Code	1
2	SP	PCH	0
3	SP-1	PCL	0
4	SP-2	PSW	0

5	\$FFDE	AAL	1
6	\$FFDF	AAH	1
7	??	I0	?
8	??	I0	?
(1)	new PC	Op Code	1

* WTF with all the I0 cycles?

* Order of reading new addr and pushing old addr may be wrong.

33a RET

(RET)

(1 byte)

(5 cycles)

1	PC	Op Code	1
2	SP	PCL	0
3	SP+1	PCH	0
4	??	I0	?
5	??	I0	?
(1)	new PC	Op Code	1

* WTF with all the I0 cycles?

33b RET1

(RET1)

(1 byte)

(6 cycles)

1	PC	Op Code	1
2	SP	PSW	0
3	SP+1	PCL	0
4	SP+2	PCH	0
5	??	I0	?
6	??	I0	?
(1)	new PC	Op Code	1

* WTF with all the I0 cycles?